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Abstract

Abstract

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Denotation

HPC	高性能計算 (High Performance Computing)

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Chapter 1 Introduction

1.1 The Standard Model

The Standard Model of particle physics is the theoretical framework that describes the known elementary particles and their interactions: the electromagnetic, weak, and strong interactions. It has successfully described a good range of phenomena observed in particle physics experiments.

In the Standard Model, matter is composed of elementary spin-1/2 fermions, which are divided into quarks and leptons. These particles are arranged into three generations with progressively larger masses. The six quarks are the up, down, charm, strange, top, and bottom quarks, while the six leptons are the electron, muon, tau, and their corresponding neutrinos. Ordinary matter is composed primarily of particles belonging to the first generation, whereas the heavier generations are unstable and are generally produced only in high-energy processes.

The interactions between these particles are mediated by spin-1 gauge bosons. The electromagnetic interaction is mediated by the photon, the weak interaction by the $W^{\pm}$ and Z bosons, and the strong interaction by gluons. Mathematically, these interactions are described by the gauge symmetry

$$SU(3)_C \times SU(2)_L \times U(1)_Y,$$

where $SU(3)_C$ describes quantum chromodynamics (QCD), while $SU(2)_L \times U(1)_Y$ de-

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scribes the electroweak interaction. EM? A crucial component of the Standard Model is the Higgs mechanism. The electroweak gauge symmetry is spontaneously broken when the Higgs field acquires a nonzero vaccum expectation value, allowing the W and Z bosons to acquire mass while leaving the photon massless. Through Yukawa interactions with the Higgs field, the charged fermions also acquire their masses. The physical excitation of the Higgs field is the Higgs boson.

The Standard Model has achieved remarkable experimental success. Measurements of electromagnetic, weak, and strong process over a wide range of energy scales have shown excellent agreement with its predictions. The discoveries of the W and Z bosons, the top quark, and finally the Higgs boson provided major experimental confirmations of central elements of the theory. In particular, the observation of a Higgs boson with a mass of approximately 125 GeV by the ATLAS and CMS experiments completed the set of elementary particles predicted by the Standard Model. Precision measurements performed at previous colliders and at the CERN Large Hadron Collider (LHC) have continued to test the theory with increasing accuracy, and so far no conclusive deviation from the Standard Model has been observed.

1.2 Limitation of the Standard Model

Nevertheless, despite its extraordinary success, the Standard Model still cannot provide a complete description of nature. Several experimental observations and theoretical questions remain unexplained within its framework.

The Standard Model provides no particle candidate for dark matter, although astronomical and cosmological observations strongly indicate that dark matter constitutes a

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substantial fraction of the matter content of the Universe. Similarly, the observed predominance of matter over antimatter cannot be fully explained.

hierarchy problem. gravity

Of particular relevance to this analysis is the pattern of fermion generations and masses. The Standard Model contains three generations of quarks and leptons, but it does not explain why there are three generation or why they have their masses as they are. The existence of repeated generations naturally raises the question of whether quarks and leptons are truly elementary particles.

Historically, objects once regarded as fundamental have repeatedly found to have internal structure. Atoms were shown to consist of nuclei and electrons, nuclei were found to contain protons and neutrons, and protons and neutrons were eventually understood as bound states of quarks and gluons. This historical progression motivates the possibility that quarks and leptons may themselves possess a deeper substructure.

1.3 Compositeness Model

One class of theories exploring this possibility is known as compositeness models. In these models, quarks and leptons are not elementary but are bound states of more fundamental constituents, called preons. The binding interaction is assumed to become strong at a characteristic compositeness scale, Λ. In analogy with other composite quantum systems, such a structure would naturally allow excited states of the observed fermions. Therefore, the existence of excited quarks would be direct evidence that quarks are composite rather than elementary. The LHC, with proton-proton collisions at multi-TeV energies, provides a powerful environment to search for these states.

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For transitions between an excited quark q^* and an ordinary quark q, the interaction with the Standard Model gauge fields can be written as this effective Lagrangian

$$\mathcal{L}_{\text{eff}} = \frac{1}{2\Lambda}\bar{q}_R^*\sigma^{\mu\nu}\left[g_s f_s \frac{\lambda_a}{2}G_{\mu\nu}^a + gf\frac{\boldsymbol{\tau}}{2}W_{\mu\nu} + g'f'\frac{Y}{2}B_{\mu\nu}\right]q_L + \text{h.c.}$$

where Λ denotes the compositeness scale. The field-strength tensors $G_{\mu\nu}^a$, $W_{\mu\nu}$, and $B_{\mu\nu}$ correspond to the $SU(3)$, $SU(2)$ and $U(1)$ gauge fields, respectively, while g_s, g and g' are the corresponding Standard Model gauge couplings. The dimensionless parameters f_s, f, and f' characterize the strength of the couplings. They are generally taken to be of order unity.

The strong-interaction term proportional to f_s give rise to the q^*qg vertex, while the electroweak terms generate couplings of the excited quark to $q\gamma$, qZ, and qW. In particular, the effective coupling to a photon is

$$f_\gamma = fT_3 + f'\frac{Y}{2},$$

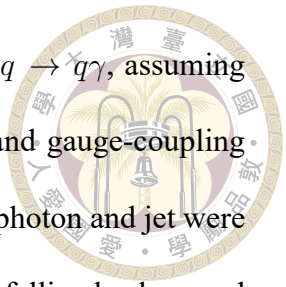
where T_3 is the third component of weak isospin and Y is the hypercharge. The corresponding radiative decay width is proportional to f_γ^2. The couplings are taken to be equal in this analysis,

$$f_s = f = f'.$$

1.4 Motivaiton for the search for excited charm quark

The CMS and the ATLAS collaboration has previously search for excited quarks. ATLAS performed a search in the γ + jet final state using 3.2 fb^{-1} of proton-proton collision data collected at the center-of-mass energy of $\sqrt{s} = 13$ TeV. The search considered

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excited quarks produce through quark-gluon fusion and decaying as $q \to q\gamma$, assuming a compositeness scale equal to the excited-quark mass, $\Lambda = M_{q^*}$, and gauge-coupling multipliers $f_s = f = f' = 1$. Events containing an energetic isolated photon and jet were selected, and the resulting $m_{\gamma j}$ distribution was fitted with a smoothly falling background function to search for a resonant excess. No significant deviation from the Standard Model prediction was observed, and excited-quark masses below 4.4 TeV were excluded at 95% confidence level.

The CMS Collaboration has also performed searches for excited quarks in the $\gamma+$jet final state at $\sqrt{s} = 13$ TeV. In addition to an inclusive search for excited light-flavor quarksm CMS considered the production of excited bottom quarks,

$$bg \to b^* \to b\gamma,$$

where identification of the bottom-flavored jet was used to enhance the sensitivity to the b^* signal. No significant excess was observed. For the coupling $f = 1.0$, excited light-flavor quarks were excluded for mass below 6.0 TeV while excited bottom-quark was excluded for masses below 2.2 TeV.

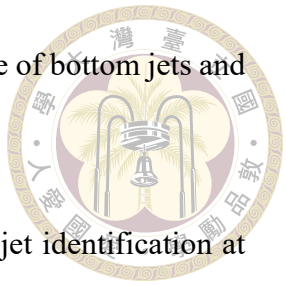
This analysis extends the same analysis strategy to the search for an excited charm quark, which would target

$$cg \to c^* \to c\gamma,$$

and would therefore require the identification of the final-state jet ad originating from a charm quark. Historically, this has been more challenging than bottom-jet identification. Charm hadrons have finite lifetimes and can produce displaced tracks and secondary vertices, but their masses and lifetimes are smaller than those of bottom hadrons. As a result,

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the experimental characteristics of charm jets tend to lie between those of bottom jets and jets originating from light-flavor quarks.



Significance progress has nevertheless been achieved in charm-jet identification at CMS. Modern heavy-flavor tagging algorithms, including DeepCSV and DeepJet, use multivariate information from charged and neutral particle-flow candidates, reconstructed tracks, and secondary vertices to estimate the probability that a jet originates from a given quark flavor. For charm identification, two complementary discriminators are defined: CvsL, which separates charm jets from light-flavor and gluon jets, and CveB, which separates charm jets from bottom jets.

The DeepJet algorithm represents an important imporvement in this capability. Unlike earlier algorithms based on a limited set of reconstructed quantities, DeepJet employs a deep neural-network architecture with information from particle-flow candidates and secondary vertices. Studies using simulated 2017 CMS events show that DeepJet provides lower misidentification rates than DeepCSV for both light-flavor and bottom jets at a given charm-tagging efficiency.

The development provides the experimental basis for extending previous excited-quark searches to the charm-quark sector. The search presented in this thesis follows the established $\gamma +$ jet resonance strategy used by ATLAS and CMS while specifically targeting the excited charm-quark hypothesis. The final state is characterized by an energetic photon and a charm-tagged jet, and the signal is searched for as a resonant enhancement in their invariant-mass distribution.

This analysis uses proton-proton collision data collected by the CMS experiment in 2017 at a center-of-mass energy of $\sqrt{s} = 13$ TeV, corresponding to an integrate luminos-

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ity of 41.8 fb^{-1}. Candidate events are selected by requiring a high-energy photon together with a charm-tagged jet, and the photon-jet invariant mass, $m_{\gamma j}$, is used as the primary observable for the resonance search. The dominant Standard Model background is expected to produce a smoothly falling $m_{\gamma j}$ spectrum. Rather than relying on simulated samples to predict its normalization and shape, the background is determined from data using a data-driven parametric model. A possible c^* signal would therefore appear as a localized excess over the background distribution near the hypothetical excited charm-quark mass.

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Chapter 2 Experimental setup

2.1 The Large Hadron Collider

The Large Hadron Collider (LHC) is a circular particle accelerator located at the European Organization for Nuclear Research (CERN), near Geneva, Switzwerland. With a circumference of approximately 27 km, it is the world's largest and highest-energy particle collider. The LHC was designed to provide proton-proton and heavy-ion collisions at unprecedented center-of-mass energies and luminosities, enabling detailed studies of the Standard Model (SM) and searches for phenomena beyond the Standard Model. During Run 2, which took place from 2015 to 2018, proton beams were accelerated to an energy of 6.5 TeV per beam, corresponding to a center-of-,ass energy of $\sqrt{s} = 13$ TeV. The data analyzed in this analysis were collected by the CMS experiment during the 2017 proton-proton collision run at this center-of-mass energy.

2.1.1 CERN Accelerator Complex

Before entering the LHC, protons pass through a sequence of smaller accelerators that progressively increase their energy. For the 2017 data-taking period, protons were initially obtained by ionizing hydrogen atoms and were accelerated by the Linear Accelerator 2. The beam was subsequently injected into the Proton Synchrotron Booster (PSB), followed by the Proton Synchrotron (PS) and the Super Proton Synchrotron (SPS). In the SPS, the protons were accelerated to an energy of 450 GeV before being transferred to the LHC.

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This accelerator chain allows the beam intensity, energy, and bunch structure required for proton-proton collisions to be established before injection into the main LHC ring.

Inside the LHC, two proton beams circulate in opposite directions in separate ultra-high-vacuum beam pipes. The beams are accelerated by radio-frequency cavities, while superconducting magnets guide and focus the particles along their circular trajectories. During Run 2, each beam reached an energy of 6.5 TeV. The main dipole magnets operate at cryogenic temperatures of approximately 1.9 K, allowing the superconducting magnets to generate the strong magnetic fields required to bend the high-energy proton beams around the LHC ring.

The proton beams are not continuous but are organized into bunches. During normal LHC operation, consecutive bunches can be separated by 25 ns, corresponding to a bunch-crossing frequency of approximately 40 MHz. At selected interaction points around the ring, the two counter-rotating beams are brought into collision. Four major experiments located at these interaction points are: ATLAS, CMS, ALICE, and LHCb.

2.1.2 Luminosity

In addition to the collision energy, an important parameter characterizing the performance of a collider is its instantaneous luminosity, $\mathcal{L}$. The event production rate for a physical process with cross section σ is given by

$$\frac{dN}{dt} = \mathcal{L}\sigma,$$

where dN/dt is the rate at which the process occurs. A higher instantaneous luminosity therefore increases the number of collisions available for studying rare process. The total

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amount of data collected during the period is characterized by the integrated luminosity,

$$\mathcal{L}_{\text{int}} = \int \mathcal{L}(t)dt.$$

For a process with production cross section σ, the expected number of produced events is approximately

$$N = \sigma \times \mathcal{L}_{\text{int}},$$

before accounting for detector acceptance and event-selection efficiencies. The 2017 proton-proton collision data set used in this analysis correspond to an integrated luminosity of 41.8 fb^{-1}, and these collision events were recorded by the Compact Muon Solenoid (CMS) experiment.

2.2 The CMS experiment

!!gap between the barrel and endcap region!!

The Compact Muon Solenoid (CMS) detector sits at one of the collision points of LHC. It is a general-purpose detector and serves to observe possible new physics phenomena occurring in the LHC. It is designed to reconstruct particles produced in proton-proton collisions. By measuring the trajectories, momenta, and energies of particles emerging from each collision, information from the different detector components can be combined to reconstruct the underlying collision event.

The CMS detector has a cylindrical structure consisting of several concentric detector layers surrounding the interaction point. It has an overall length of approximately 21 m, a diameter of about 15 m, and a total mass of approximately 14000 tonnes. The detector is arranged in a central cylindrical barrel together with two endcap regions. Its supercon-

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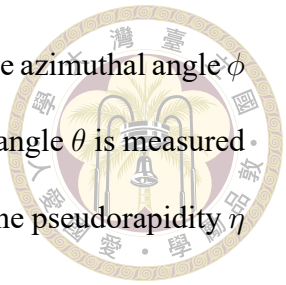
ducting solenoid, which surrounds the inner tracking system and calorimeters, produces a magnetic field of approximately 4 T. Charged particles produced in a collision follow curved trajectories in this magnetic field. The direction of the curvature allows the sign of the particle charge to be determined, while the amount of curvature provides a measurement of its momentum. Particles with higher transverse momentum exhibit less curvature than particles with lower transverse momentum. The magnetic flux is returned through a massive steel return yoke surrounding the solenoid, within which the muon detectors are installed.

Starting from the interaction point and moving radially outward, the principal components of CMS are the silicon tracking system, the electromagnetic calorimeter (ECAL). the hadron calorimeter (HCAL), the superconducting solenoid, and the muon detection system. The silicon tracker provides precise measurements of the trajectories of charged particles. The ECAL measures the energies of electrons and photons, while the HCAL measures the energy deposited by hadrons. Muons generally pass through both calorimeters and are subsequently detected by muon chambers located outside the solenoid and interleave with the steel returned yoke.

2.2.1 CMS Coordinate System

The CMS experiment uses a right-handed Cartesian coordinate system with its origin located at the nominal proton-proton interaction point at the center of the detector. The x-axis points radially inward toward the center of the LHC ring, the y-axis points vertically upward, and the z-axis is defined along the direction of the counterclockwise beam. The $x-y$ plane therefore corrseponds to the plane transverse to the beam direction.

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Particle directions are described using cylindrical coordinates. The azimuthal angle ϕ is measured in the transverse plane around the z-axis, while the polar angle θ is measured with respect to the positive z-axis. Rather than the polar angle itself, the pseudorapidity η is generally used and is defined as

$$\eta = -\ln\left[\tan\left(\frac{\theta}{2}\right)\right].$$

The central region of the detector, corresponding to trajectories approximately perpendicular to the beam axis, is characterized by values of η close to zero. Increasing values of $|\eta|$ correspond to increasingly forward directions, closer to the beam axis.

The momentum component perpendicular to the beam direction is referred to as the transverse momentum and is defined as

$$p_{\mathrm{T}} = \sqrt{p_x^2 + p_y^2}.$$

Similarly, the transverse energy E_{T} is defined with respect to the transverse plane.

The angular separation bewteen two reconstructed objects is conventionally expressed in the η-ϕ plane as

$$\Delta R = \sqrt{(\Delta\eta)^2 + (\Delta\phi)^2},$$

where $\Delta\eta$ and $\Delta\phi$ are the differences in pseudorapidity and azimuthal angle between the two objects. This quantity is widely used in particle reconstruction and event selection to characterize the spatial separation of reconstructed particles.

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2.2.2 Silicon Tracker



The silicon tracker is in the innermost component of CMS and surrounds the interaction points inside the superconducting solenoid. Its primary purpose is to reconstruct the trajectories of charged particles produced in proton-proton collisions. As charge particles traverse the silicon sensors, they deposit energy through ionization and produce localized signals, or hits. By combining hits recorded in successive detector layers, the trajectories of charged particles can be reconstructed with high precision. The curvature of these trajectories in the magnetic field then provides a measurement of the particle momentum and charge.

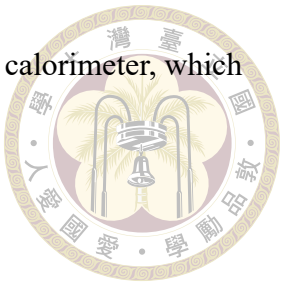
During the Run 2 data-taking period relevant to this analysis, the CMS tracking system consisted of two complementary silicon detector technologies: a silicon pixel detector close to the interaction point and a silicon strip detector surrounding it. The pixel detector provides high-granularity measurements close to the beam line, where the particle density is the greatest, while the strip detector extends the tracking volume to large radii.

The pixel detector is for accurately determining the positions of interaction and decay vertices. Its fine spatial segmentation enables precise measurements close to the proton-proton collision point and allows tracks originating from secondary vertices to be distinguished from those produced at the primary vertex. Such information is essential for identifying jets originating form heavy-flavor quarks, whose hadrons typically travel a measurable distance before decaying.

Outside the pixel detector, the transverse momentum of a charged particle can be inferred form the curvature of its reconstructed trajectory in the silicon strip tracker.

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Immediately outside the tracking system lies the electromagnetic calorimeter, which provides precise measurements of the energies and photons.



2.2.3 Electromagnetic Calorimeter

The electromagnetic Calorimeter (ECAL) surrounds the silicon tracker and is also located inside the CMS superconducting solenoid. It is to measure the energies and positions of electrons and photons produced in proton-proton collisions. Since photons do not leave tracks in the silicon tracker, the ECAL provides the principal measurement of their energy.

The CMS ECAL is a homogeneous calorimeter constructed from lead tungstate ($PbWO_4$)) scintillating crystals. When an energetic electron or photon enters the ECAL, it initiates an electromagnetic shower. The resulting shower particles deposit their energy in the crystals, producing scintillation light. The amount of light collected is proportional to the deposite energy and is used to reconstruct the energy of the incident particle.

The ECAL is divided into a central barrel region (EB) and two endcap regions (EE). The barrel is composed of crystals arranged radially around the interaction point, while the endcaps extend the coverage into the forward regions. In the transition region between the barrel and endcaps, the detector geometry and the amount of material in front of the calorimeter lead to reduced reconstruction performance, and this region is therefore commonly excluded from measurements.

The high density and short radiation length of $PbWO_4$ allow electromagnetic showers to be contained within a compact detector volume. At the same time, the fine transverse segmentation of the crystal array provides precise spatial information about the energy

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deposits. These properties enable the ECAL to achieve the high energy resolution.

Before reaching the ECAL, electrons may lose energy through bremsstrahlung, while photons may convert into an e^+e^- pair. In the magnetic field of CMS, the resulting charged particles are deflected predominantly in the azimuthal direction. Consequently, the electromagnetic shower energy may be distributed over several neighboring crystals rather than being contained in a single crystal. To account for this effect, energy deposits in adjacent ECAL crystals are first combined into clusters. Related clusters are subsequently grouped into larger structures known as *superclusters*, which are designed to recover energy spread by bremsstrahlung and photon conversions. The supercluster energy, after applying the appropriate calibrations and corrections, provides the basis for reconstructing the energy of electron and photon candidates.

Photon identification additionally exploits the spatial distribution of energy within the ECAL. One commonly used quantity is R_9, defined as the ratio of the energy contained in a 3×3 array of crystals centered on the most energetic crystal to the total energy of the supercluster.

Unconverted photons generally deposit a large fraction of their energy in a compact group of crystals and consequently tend to have large values of R_9, whereas converted photons often produce a broader energy distribution.

2.2.4 Hadron Calorimeter

The hadron calorimeter (HCAL) surrounds the ECAL and is designed to measure the energies of hadrons produced in proton-proton collisions. While electrons and photons deposit most of their energy in the ECAL, hadrons generally penetrate further into the

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detector and interact with the material of the HECAL through strong interactions.

Unlike the homogeneous crystal design of the ECAL, the CMS HCAL is a sampling calorimeter. It consists of alternating layers of dense absorber material and active scintillator. When an energetic hadron interacts with the absorber, it produces a cascade of secondary particles known as a hadronic shower. The charged components of the shower pass through the scintillator layers and generate light, which collected and converted into an electrical signal. The measured signal is then used to estimate the energy deposited by the incident particle.

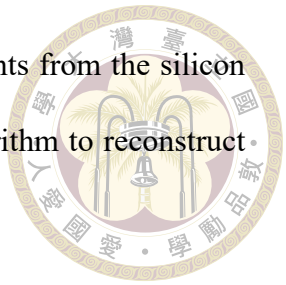
The HCAL is divided into several regions in order to provide broad angular coverage. The hadron barrel (HB) surrounds the ECAL barrel, while the endcap (HE) extends the coverage into larger values of pseudorapidity. An additional hadron outer (HO) calorimeter is located outside the solenoid in the barrel region and provides supplementary sampling of hadronic showers that extend beyond the main HCAL volume. At large pseudorapidities, the hadron forward (HF) calorimeters provide coverage close to the beam direction.

The HB and HE calorimeters are located inside the CMS superconducting solenoid. Because the available radial space inside the magnet is limited, some high-energy hadronic showers may not be fully contained within these calorimeters. The HO detector provides additional measurements outside the solenoid and improves the detection of energy leaking through the inner calorimeters.

HCAL plays an important role in the reconstruction of jets. A jet originates from the fragmentation and hadronization of a quark or gluon and therefore consists of many particles distributed over a angular region. The energy deposits produced by these particles

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in the ECAL and HCAL, together with charged-particle measurements from the silicon tracker, are combined by the CMS particle-flow reconstruction algorithm to reconstruct the particles forming the jet.



2.2.5 Muon System

The muon system forms the outermost major detection system of CMS and is designed to identify and measure muons with high efficiency and precision. Muons are considerably more penetrating than electrons and hadrons and typically pass through both the ECAL and HCAL with relatively little energy loss. Dedicated muon detectors are therefore installed outside the calorimeters, within and around the steel return yoke of the CMS solenoid.

The CMS muon system is arranged in both barrel and endcap regions. In the barrel, drift-tube chambers are used to reconstruct muon trajectories, while cathod-strip chambers provide measurements in the endcap regions. Resistive-plate chambers are additionally used throughout much of the muon system to provide fast timing information and contribute to the muon trigger.

As a muon traverse the detector, its trajectory is measured independently in both the inner tracker and the external muon chambers. The strong magnetic field generated by the CMS solenoid bends the trajectory of the charged muon, while the steel return yoke also provides a return path for the magnetic flux.

Muon candidates can be reconstructed using information from the muon chambers alone or by matching muon-system measurements to tracks reconstructed in the silicon tracker. The combination of these detector systems allows CMS to distinguish muons from

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other charged particles and to reconstruct their momenta over a wide kinematic range.



2.2.6 Trigger and Data Acquisition System

The high collision rate of the LHC produces far more events than can be permanently recorded and processed. Proton bunches cross at intervals of 25 ns, corresponding to a bunch-crossing frequency of approximately 40 MHz. As a result, the CMS detector must rapidly identify potentially interesting collision events while rejecting the large majority of events that are not suitable for analyses. This task is performed by the CMS trigger and data acquisition system.

CMS employs a two-level trigger architecture consisting of the Level-1 (L1) trigger and the High-Level Trigger (HLT). The L1 trigger is implemented using dedicated hardware and performs a fast initial selection using information from the calorimeters and muon detectors. It searches for signatures such as energetic electrons, photons, jets, or muons, as well as combinations of such objects. The L1 system reduces the event rate from the extremely large collision rate to approximately 100 kHz.

Events accepted by the L1 trigger are transferred to the High-Level Trigger, which is implemented in software and executed on a large computing farm. In contrast to the relatively simle information available at L1, the HLT has access to more complete detector information and can perform reconstruction algorithms similar to those used in offline analysis.

In this analysis, events are selected using a high-transverse-momentum photon trigger. For the 2017 data set, the trigger `HLT_Photon200` is used. The offline photon transverse-momentum requirement is chosen above the trigger threshold in order to op-

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erate in the region where the trigger efficiency is approximately constant and close to its maximum value.

After an event is accepted by the trigger system, information from the detector is assembled by the data acquisition system and stored for further reconstruction and analysis. These data are subsequently processed and distributed using the Worldwide LHC Computing Grid (WLCG), which provides the computing and storage infrastructure

The computing infrastructure is organized into a hierarchical system of computing centers. The Tier-0 center at CERN receives the data recorded by the CMS detector and performs the initial processing and reconstruction of the collision events. Then the data are distributed to large Tier-1 computing centers.

The Tier-1 centers provide substantial storage and computing resources and perform additional data reconstruction using updated detector calibration and alignment information. Tier-1 centers are distributed across several countries, including France, Germany, Italy, Spain, Taiwan, the United Kingdom, and the United States. Taiwan therefore plays a direct role in the worldwide computing infrastructure supporting the CMS experiment.

The Tier-1 centers further distribute data to Tier-2 facilities, which provide computing resources primarily for physics analysis and simulation.

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Chapter 3 Object Reconstruction

3.1 Object Reconstruction

Event reconstruction at the CMS experiment uses the particle flow (PF) algorithm to reconstruct and identify individual particles in each event, including photons, electrons, muons, and charged and neutral hadrons (collectively known as PF candidates). This approach combines information from all CMS subdetectors to achieve optimal event reconstruction.

Photon energies are measured using the electromagnetic calorimeter (ECAL) detector. For electrons, the energy is determined by combining the track momentum at the primary vertex, the corresponding ECAL cluster energy, and the summed energy of bremsstrahlung photons emitted along the trajectory. Muon momentum is obtained from the curvature of their tracks in the magnetic field. Charged hadrons are reconstructed using tracker information together with ECAL and HCAL energy deposits, while neutral hadron energies are derived from the corrected ECAL and HCAL measurements.

3.2 Photon Reconstruction

The CMS electromagnetic calorimeter (ECAL) is designed to provide excellent energy resolution and high granularity, enabling precise photon measurements at the LHC. Since photons do not leave tracks in the inner tracker, their energy is measured entirely using the EACL. As photons traverse the detector material, they may convert into an e^+e^-

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pair, initiating an electromagnetic shower. The resulting secondary particles deposit their energy across multiple ECAL crystals, forming localized clusters.

Due to the combined influence of secondary interactions in the tracker material - including bremsstrahlung and photon conversions - along with the 3.8 T solenoidal magnetic field of CMS, these clusters become spread in the azimuthal direction. To fully capture the shower development, clusters are combined using dedicated algorithms to form superslucters, which are used to reconstruct photon candidates. After applying energy corrections, the photon momentum is assigned based on the supercluster energy and the position of the reconstructed primary vertex.

Photons are classified as converted or unconverted, depending on whether they undergo conversion in the tracker material. This is determined using the variable $R9$, defined as the ratio of the energy contained within a 3×3 crystal array centered on the seed crystal to the total supercluster energy. In the ECAL barrel region, photons with $R9 > 0.94$ are considered unconverted, and their energy is taken from the 5×5 crystal matrix surrounding the highest energy crystal. Otherwise, the photon is classified as converted, and the full supercluster energy is used.

At this reconstruction stage, not all photon candidates are genuine photons, as several processes can mimic photon signatures. Therefore, additional identification criteria are applied to select high-quality photon candidates for analysis.

3.3 Jet Reconstruction

Jets are reconstructed as collimated sprays of particles resulting from the hdronization of quarks and gluons, depositing energy in the CMS calorimeters through electromagnetic

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and hadronic interactions. Several jet reconstruction algoritms are used in CMS. PF Jets are used in this analysis, combine information from all CMS subdetectors to reconstruct individual particles, providing improved momentum and spatial resolution.

Jets are reconstructed using the ant-k_T clustering algorithm with a distance parameter $R = 0.4$ in the $y - \phi$ space. The Particle-Flow algorithm identifies and reconstructs all stable particles in the event by combining information from the tracking system, calorimeters, and muon detectors. To mitigate the effects of pileup, PF jets can be further categorized according to the pileup mitigation technique applied. In this analysis, PF jets reconstructed with the Charged Hadron Substraction (CHS) method are used. In the CHS approach, charged particles originating from pileup vertices are removed prior to jet clustering, thereby reducing contamination from additional interactions.

In addition, jet identification criteria are applied to suppress fake jets arising from detector noise and non-collision backgrounds, such as electronic noise or beam-related effects. These criteria ensure that only well-reconstructed jets are selected for further analysis.

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Chapter 4 Analysis Strategy

This analysis searches for the production of an excited charm quark, c^*, in proton-proton collision at a center-of-mass energy of $\sqrt{s} = 13\,\text{TeV}$. The signal process considered is the gauge-mediated production and decay $cg \to c^* \to c\gamma$, which gives rise to a final state containing a photon and a charm-quark jet, referred to as a $\gamma + c$-jet final state. Since quarks are not observed directly in the detector, the charm quark is reconstructed as a jet and identified using charm-tagging discriminators.

4.1 Dataset

The search is performed using the 2017 proton-proton collision data set collected by the CMS detector, corresponding to an integrated luminosity of $41.8\,\text{fb}^{-1}$. The data set is list in Table 4.1

Data Sample
/SinglePhoton/Run2017B-UL2017_MiniAODv2_NanoAODv9-v1/NANOAOD
/SinglePhoton/Run2017C-UL2017_MiniAODv2_NanoAODv9-v1/NANOAOD
/SinglePhoton/Run2017D-UL2017_MiniAODv2_NanoAODv9-v1/NANOAOD
/SinglePhoton/Run2017E-UL2017_MiniAODv2_NanoAODv9-v1/NANOAOD
/SinglePhoton/Run2017F-UL2017_MiniAODv2_NanoAODv9-v1/NANOAOD

Table 4.1: List of 2017 CMS dataset used in the analysis

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4.2 Trigger

4.3 Object Reconstruction and Event Selection

4.3.1 Photon Selection

Each candidate event is required to contain at least one photon with transverse momentum $p_T^{\gamma} > 240$ GeV, located within the ECAL barrel region ($|\eta^{\gamma}| < 1.44$). Photon candidates must satisfy the medium working point of the EGamma POG recommended cut-based Photon ID criteria [REFERENCE], which suppresses background from jets misidentified as photon while maintaining high efficiency.

In addition, an electron veto is applied to reject electrons misidentified as photons by rejecting photon candidates whose associated supercluster is matched to a reconstructed electron track with no missing hits in the innermost tracker layers.

4.3.2 Jet Selection

Each candidate event is also required to contain at least one jet with transverse momentum $p_T^{Jet} > 170$ GeV and pseudorapidity $|\eta^{\gamma}| < 2.4$. Jets must satisfy the tight jet identification criteria recommended by JetMET POG[REFERENCE]. Charm-tagging requirements are applied using the DeepJet algorithm. Jets passing the medium working point of the DeepJet charm discriminator are considered as c-jets [REFERNECE].

Finally, since particle-flow jets are reconstructed using the anti-k_T clustering algorithm with a distance parameter $R = 0.4$[REFERNECE], a separation requirement of $\Delta R(\gamma, jet) > 0.4$ is imposed to ensure that the photon and jet are well-separated.

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If more than one photon or jet candidates present in an event, the leading objects in transverse momentum are selected for further analysis.

TABLES: P65 Selection efficiency (in for SM Coupling f = 1.0, 0.5, 0.1

4.4 Signal modeling

4.4.1 Signal Sample Generation

Signal samples of excited charm-quark production were generated in pp collision at a center-of-mass energy of $\sqrt{s} = 13$ TeV. The signal process is the resonance production of an excited charm quark through charm-gluon fusion, followed by its decay into a charm quark and a photon:$cg \to c^* \to c\gamma$.

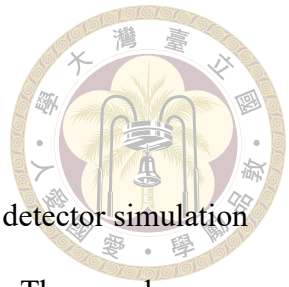
The events were generated at leading order using the excited-fermion model implemented in Pythia 8. The excited charm quark is represented by POG identifier 4000004. The three coupling parameters are assumed to have identical values: $f_s = f = f' \equiv f$, and signal samples are generated at c^* mass points from 1.0 to 3.0, TeV for three coupling values: $f = 0.1, 0.5$, and 1.0.

The compositeness scale was set equal to the excited charm-quark mass for every sample: $\Lambda = M_{c^*}$. The generated excited charm quark was forced to decay through $c^* \to c\gamma$ by disabling its other decay modes.

Parton showering, hadronization, multiparton interactions, and the underlying event were simulated with Pythia 8 using the CUETP8M1 tune.

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4.4.2 Detector Simulation and Event Reconstruction



The generated particles were passed through the complete CMS detector simulation using GEANT4 10.4.3, as integrated into the CMS Software framework. The samples were produced with CMSSW 10_6_30 using `106X_mc2017_realistic_v6` conditions, the Run2_2017 era, the CMS extended detector geometry, and the Realistic25ns13TeVEarly2017Collision beam-spot configuration.

The complete processing sequence was

$$\text{GEN-SIM} \rightarrow \text{DIGI-RAW-HLT} \rightarrow \text{AODSIM} \rightarrow \text{MINIAODSIM} \rightarrow \text{NANOAODSIM}.$$

During the GEN-SIM stage, Pythia generated the hard-scattering event, parton shower, and hadronisation, while GEANT4 simulated the CMS detector.

Detector digitization included additional proton–proton interactions using the 2017 Ultra Legacy pileup scenario, `2017_25ns_UltraLegacy_PoissonOOTPU`. The digitized events were subsequently processed through the Level-1 trigger, the 2017 high-level trigger configuration, and the standard CMS reconstruction algorithms. The reconstructed events were converted to MINIAODSIM and finally to NANOAODSIM for use in the analysis.

4.4.3 Signal Modeling

The four-momenta of the selected photon and jet are combined to obtain the invariant mass distribution of c^*, and this distribution is modeled using a double-sidede Crystal Ball function. The function consists of a Gaussian core and power-law tails on both sides of

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the peak. The fits are performed for every simulated mass and coupling, and the simulated events are weighted using the nominal pileup, photon-identification, and charm-tagging weights. The fitted shapes are shown in [combined$_signal_shapes$].



The normalization of each signal model is obtained from the expected signal yield,

$$N_{\text{sig}} = \sigma(pp \to c^*)\mathcal{B}(c^* \to c\gamma)\mathcal{L}A\epsilon, \tag{4.1}$$

where σ is the production cross section, $\mathcal{B}$ is the branching fraction, $\mathcal{L}$ is the integrated luminosity, and $A\epsilon$ is the accaptance $\times$ event-selection efficiency.

as the mass increase, why the shape become wider as the coupling increase, why the shape become wider

4.5 Background Modeling

$$w_i^{\text{raw}} = \frac{\mathcal{L}\sigma}{\sum_j w_j^{\text{gen}}} \times w_i^{\text{gen}}$$

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Chapter 5 Systematics Uncertainties

5.1 Integrated Luminosity

The signal normalization is proportional to the integrated luminosity of the data sample. Therefore, the uncertainty in the measured integrated luminosity directly affects the expected signal yield. This uncertainty is treated as a normalization uncertainty.

For the 2017 data-taking period, the integrated luminosity used in this analysis is

$$\mathcal{L} = 41.8 \text{ fb}^{-1}.$$

The corresponding luminosity is taken as 2.3%. The nominal signal yield is computed using the central value of the integrated luminosity,

$$N_{\text{sig}} = \sigma_{\text{sig}} \, \mathcal{L} \, A\epsilon,$$

where σ_{sig} is the signal production cross section and $A\epsilon$ is the acceptance times selection efficiency. The luminosity-up and luminosity-down variations are obtained by scaling the nominal signal yield by

$$N_{\text{sig}}^{\text{LumiUp}} = N_{\text{sig}}(1 + 0.023), \qquad N_{\text{sig}}^{\text{LumiDown}} = N_{\text{sig}}(1 - 0.023).$$

Since the luminosity uncertainty changes only the overall normalization, the nominal invariant-mass distribution is kept unchanged. In the statistical model, this uncertainty

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is implemented as a rate uncertainty on the signal process, with a log-normal nuisance parameter corresponding to a relative uncertainty of 2.3%.



5.2 Pileup

[1] In proton-proton collisions, more than one interaction can occur in the same bunch crossing. In addition to the hard-scattering process of interest, additional soft interactions are also recorded in the event. This is called pileup. Since the pileup distribution in simulated sample does not exactly match that observed in data, a pileup reweighting is applied to simulated events.

The pileup distribution in simulation is compared with the pileup profile estimated in data using the number of true interactions. The nominal pileup profile in data is obtained using the central value of the assumed inelastic proton-proton cross section. Two additional profiles are produced by varying this cross section upward and downward. In this analysis, the nominal pileup profile corresponds to an inelastic cross section of 69.2 mb, while the upward and downward variations correspond to 72.4 mb and 66.0 mb, respectively.

A nominal pileup weight is calculated as the ratio of the normalized pileup distribution in data to that in simulation. The upward and downward pileup weights are obtained in the same way, using the corresponding varied data pileup profiles. The weights are applied event by event to correct the simulated pileup profile to match the observed data.

The central signal prediction is obtained using the nominal pileup weights. The pileup systematic variations are obtained by replacing the nominal pileup weights with the upward and downward weights, while keeping the event selection and reconstruction

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unchanged. This gives three invariant-mass distributions: the nominal distribution, the pileup-up variation, and the pileup-down variation.

The pileup uncertainty is then evaluated from the change in the final signal yield obtained with the varied pileup weights.

5.3 Energy Scale and Resolution Uncertainties

The reconstructed photon–jet invariant mass depends on the energy calibration and resolution of both the photon and the jet. The jet energy scale (JES) uncertainty accounts for uncertainties in the jet energy correction, while the jet energy resolution (JER) uncertainty accounts for the finite detector resolution in the jet energy measurement. The photon energy scale (PES) uncertainty accounts for uncertainties in the photon energy calibration. These effects change the reconstructed photon–jet invariant mass and therefore modify the signal shape.

For the jets, the nominal four-momentum is obtained by first reconstructing the raw jet momentum from the NanoAOD jet momentum and raw correction factor. The nominal jet energy correction is then applied to the raw jet transverse momentum and mass. For the 2017 simulated samples, the AK4 PUPPI jet corrections are taken from the Summer20UL17 NanoAODv15 JERC set. The nominal correction uses the L1FastJet and L2Relative correction levels, `Summer20UL17NanoV15_V1_MC_L1FastJet_AK4PFPuppi` and `Summer20UL17NanoV15_V1_MC_L2Relative_AK4PFPuppi`. The JES uncertainty is evaluated using the total JES uncertainty source, `CMS_scale_j_Total`, corresponding to `Summer20UL17NanoV15_V1_MC_Total_AK4PFPuppi`. For each jet, upward and downward JES scale factors are calculated as functions of the jet transverse momentum and

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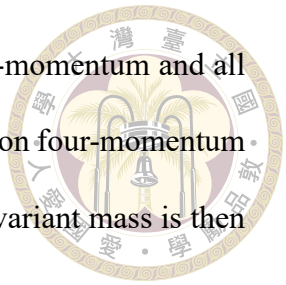
pseudorapidity. These factors are applied to both the nominally corrected jet transverse momentum and jet mass, producing JES-up and JES-down jet four-momenta.

After the nominal jet energy correction is applied, the JER smearing is applied to the corrected jet transverse momentum and mass. For the 2017 simulated samples, the JER corrections are taken from the Ultra Legacy 2017 JER set. The jet transverse momentum resolution uses `Summer19UL17_JRV3_MC_PtResolution_AK4PFPuppi`, and the corresponding JER scale factors use `Summer19UL17_JRV3_MC_ScaleFactor_AK4PFPuppi`. For each reconstructed jet, a generator-level jet match is searched for using a maximum angular distance of $\Delta R < 0.2$. If a valid generator-level match is found and satisfies the JER smearing requirements, the matched-jet method is used. Otherwise, a stochastic smearing procedure is applied. The nominal, upward, and downward JER scale factors are used to construct the nominal, JER-up, and JER-down jet four-momenta.

For the photon, the PES uncertainty is evaluated using the photon energy scale uncertainty correction from the EGM correction set. For the 2017 simulated samples, the correction is taken from `EGM_ScaleUnc.json.gz`, using the correction `UL-EGM_ScaleUnc`. For each selected photon, upward and downward photon energy scale factors are obtained from this correction and applied to the nominal photon transverse momentum, producing PES-up and PES-down photon four-momenta.

The event selection is performed using the nominal photon and the nominally corrected and smeared jet. For events passing the nominal selection, the same selected photon and jet are used to construct all variations. For the JES and JER variations, the photon four-momentum, event weights, pileup weights, and c-tagging scale factors are kept fixed at their nominal values, and only the jet four-momentum is replaced by the corresponding

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shifted four-momentum. For the PES variations, the selected jet four-momentum and all event weights are kept fixed at their nominal values, and only the photon four-momentum is replaced by the shifted photon four-momentum. The photon–jet invariant mass is then recalculated as

$$m_{\gamma j}^{\text{JESUp/Down}} = m(\gamma_{\text{nominal}} + j_{\text{JESUp/Down}}),$$

$$m_{\gamma j}^{\text{JERUp/Down}} = m(\gamma_{\text{nominal}} + j_{\text{JERUp/Down}}), \qquad m_{\gamma j}^{\text{PESUp/Down}} = m(\gamma_{\text{PESUp/Down}} + j_{\text{nominal}}).$$

This produces the nominal signal template together with JES-up, JES-down, JER-up, JER-down, PES-up, and PES-down signal templates.

The nominal and shifted signal templates are fitted with a Double-Sided Crystal Ball function. The shifted templates are not used directly as histogram morphing templates in the statistical model. Instead, the effect of each uncertainty is propagated to the fitted Double-Sided Crystal Ball parameters. For each shape parameter p, the corresponding variations are defined as

$$\Delta p_{\text{JES}} = \frac{p_{\text{JESUp}} - p_{\text{JESDown}}}{2}, \qquad \Delta p_{\text{JER}} = \frac{p_{\text{JERUp}} - p_{\text{JERDown}}}{2}, \qquad \Delta p_{\text{PES}} = \frac{p_{\text{PESUp}} - p_{\text{PESDown}}}{2}.$$

The parameter is then written in the workspace as

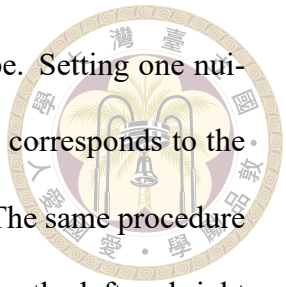
$$p(\theta_{\text{JES}}, \theta_{\text{JER}}, \theta_{\text{PES}}) = p_{\text{nominal}} + \Delta p_{\text{JES}}\theta_{\text{JES}} + \Delta p_{\text{JER}}\theta_{\text{JER}} + \Delta p_{\text{PES}}\theta_{\text{PES}},$$

where θ_{JES}, θ_{JER}, and θ_{PES} are the corresponding nuisance parameters.

For example, for the peak position x_0, the parametrization is

$$x_0(\theta_{\text{JES}}, \theta_{\text{JER}}, \theta_{\text{PES}}) = x_{0,\text{nominal}} + \Delta x_{0,\text{JES}}\theta_{\text{JES}} + \Delta x_{0,\text{JER}}\theta_{\text{JER}} + \Delta x_{0,\text{PES}}\theta_{\text{PES}}.$$

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Setting all nuisance parameters to zero gives the nominal signal shape. Setting one nuisance parameter to $+1$ or -1, while keeping the others fixed at zero, corresponds to the corresponding upward or downward shift of the fitted peak position. The same procedure is applied to the other Double-Sided Crystal Ball parameters, including the left and right widths and the tail parameters.

5.4 C-Tag Efficiency

The signal selection requires the selected jet to pass the charm-tagging requirement. The charm-tagging efficiency in simulation is corrected to match that measured in data by applying charm-tagging scale factors to simulated events. This uncertainty affects the expected signal normalization.

In this analysis, the charm-tagging scale factors are taken from the BTV correction set for the Run 2 2017 Ultra Legacy NanoAOD samples. The correction is read from `ctagging.json.gz`, using the `deepJet_wp` correction. The selected jet is required to pass the medium DeepJet charm-tagging working point, defined by

$$\text{CvB} > 0.340, \qquad \text{CvL} > 0.085.$$

For the selected jet, the scale factor is evaluated as a function of the jet hadron flavour, absolute pseudorapidity, and transverse momentum. For charm jets, corresponding to hadronFlavour $= 4$, the correction is evaluated with the `wcharm` flavour category and the medium working point. The nominal, upward, and downeard scale factors are obtained using the `central`, `up`, and `down` systematic labels, respectively.

The nominal event weight for the signal sample is multiplied by the nominal charm-

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tagging scale factor, and the upward and downward charm-tagging variations are obtained by replacing the nominal scale factor with its upward and downward variations. The event selection and reconstruction photon-jet invariant mass are kept fixed to their nominal values for these variations. Therefore, the charm-tagging uncertainty is treated as an event-weight uncertainty.

Three invariant-mass distributions are produced: the nominal distribution, the charm-tagging-up variation, and the charm-tagging-down variation. The charm-tagging uncertainty is then evaluated from the change in the final signal yield.

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Chapter 6 Results

6.1 Statistical Interpretation

To determine whether the new physics is found or not, we use the method called frequentist statiscial test [2]. This is a mathematical framework widely used in high energy physics, allowing us to treat the searching of new phenomenon as a hypothesis test. It can be used in two purposes, discovery significance and exclusion limit. In this analysis, the exclusion limit is applied, and here is the standard procedure of this measure.

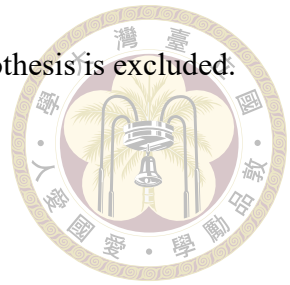
For the purposes of setting limits, the signal plus background model plays the role of H_0 (null hypothesis), and the background-only hypothesis plays the role of H_1 (the alternative hypothesis). H_0 is the hypothesis that is currently under test, we are asking whether this signal plus background hypothesis is compatible with the data. If it is not compatible to some science extent, then we reject the hypothesis.

6.1.1 P-value

By computing p-value, we are able to quantify the level of compatibility between the data and the given hypothesis. P-value is a probability, under the assumption that H is true, of obtaining data of equal or greater incompatibility with H as the observed data. If the probability is small, it means it would be rare to obtain the data this disagree with H (or this far from what H predict) as much as the observed data do, then we regard this observation as unusual assuming H is true, then we proceed to reject this hypothesis.

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Therefore, if the p-value is smaller than a specific threshold, the hypothesis is excluded.



6.1.2 Signal Strength Parameter μ

The observed $m_{\gamma j}$ distribution is described by the sum of signal and background model, using the signal-plus-background binned fit. The model is

$$\mu S(m_{\gamma j};\theta_s) + B(m_{\gamma j};\theta_b), \tag{6.1}$$

or, in terms of the expected bin yields in the i-th bin,

$$\mu s_i(\theta) + b_i(\theta). \tag{6.2}$$

Here, s_i and b_i denote the expected signal and background yields, respectively. The parameter μ is the signal strength parameter, which scales the nominal signal prediction. μ = 0 corresponds to the background-only hypothesis, while μ = 1 corresponds to the nominal signal prediction. When fitting the data with model, the fit determines the value μ that best describe the observed distribution, denoted by $\hat{\mu}$. In the procedure of limit-setting, different values of μ are tested, and the upper limit on μ is determined as the largest signal strength that still remained compatible with the observed data.

6.1.3 Likelihood Function

To quantify how well a model of given signal strength describes the data, we use the likelihood function, which is a product of Poisson distributions for all bins,

$$\mathcal{L}(\text{data}|\mu,\theta) = \prod_{i=1}^{n_{\text{bin}}} \text{P}\left(x_i|\mu s_i(\theta) + b_i(\theta)\right) \text{C}(\theta), \tag{6.3}$$

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where x_i is the observed number of event in i^{th} bin, and $C(\theta)$ is the constraint term, which contained the systematic uncertainties related nuisance parameters (θ). For a given μ, the likelihood measures how probable the observed bin yield is under the signal-plus-background model. Larger likelihood indicate better agreement between the observed data and the model.

6.1.4 Profile Likelihood Ratio

Since the likelihood depends not only on the signal-strength parameter, but also on the nuisance parameters, the compatible of data and model can be quantified using the profile likelihood ratio,

$$\lambda(\mu) = \frac{\mathcal{L}(\mu, \hat{\hat{\theta}})}{\mathcal{L}(\hat{\mu}, \hat{\theta})}. \tag{6.4}$$

In the numerator, $\hat{\hat{\theta}}$ denotes the value of θ that maximizes the likelihood for a given μ, which is the conditional maximum likelihood estimator of θ. In the denominator, $\hat{\mu}$ and $\hat{\theta}$ are the unconditional maximum likelihood estimators, $\hat{\mu}$ is the best-fit signal strength preferred by the data, and $\hat{\theta}$ is the best-fit value of the nuisance parameters. Both μ and θ are free to vary.

By definition, $0 \leq \lambda \leq 1$, with λ closer to 1 meaning the data and the value of μ are in better agreement.

In search for an excess of events, the signal contribution is considered to be positive, so μ is required to $\mu \geq 0$. However, due to the fluctuations in data, one may get a negative best fit, $\hat{\mu} \leq 0$. In such a case, the best physically allowed fit occurs for $\mu = 0$, corrsponding to the background only hypothesis. To account for this, we define

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$$\tilde{\lambda}(\mu) = \begin{cases} \dfrac{\mathcal{L}(\mu, \hat{\hat{\theta}}(\mu))}{\mathcal{L}(\hat{\mu}, \hat{\theta})} & \hat{\mu} \geq 0, \\ \dfrac{\mathcal{L}(\mu, \hat{\hat{\theta}}(\mu))}{\mathcal{L}(0, \hat{\hat{\theta}}(0))} & \hat{\mu} < 0. \end{cases} \tag{6.5}$$



When $\hat{\mu} \geq 0$, the denominator is the global maximum of the likelihood as usual. When $\hat{\mu} \leq 0$, the denominator is instead evaluated at $\mu = 0$.

6.1.5 Test Statistic

It is convenient to use the statistic

$$\tilde{t}_\mu = -2 \ln \tilde{\lambda}(\mu) = \begin{cases} -2 \ln \dfrac{\mathcal{L}(\mu, \hat{\hat{\boldsymbol{\theta}}}(\mu))}{\mathcal{L}(0, \hat{\hat{\boldsymbol{\theta}}}(0))} & \hat{\mu} < 0, \\ -2 \ln \dfrac{\mathcal{L}(\mu, \hat{\hat{\boldsymbol{\theta}}}(\mu))}{\mathcal{L}(\hat{\mu}, \hat{\boldsymbol{\theta}})} & \hat{\mu} \geq 0. \end{cases} \tag{6.6}$$

to perform the statistic test. Higher t_μ corresponds to the greater incompatibility between the data and the value of μ.

For setting an upper limit on the signal strength, a one-sided test statistic is used. The purpose of an upper-limit test is asking whether μ is too large to be compatible to the observed data, therefore, if the best-fit signal strength $\hat{\mu} \geq \mu$, the data prefer a signal contribution larger than the tested μ, this should not be used to reject the tested μ. And we then define the test statistic as

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$$\tilde{q}_\mu = \begin{cases} -2\ln \dfrac{\mathcal{L}\left(\mu, \hat{\hat{\boldsymbol{\theta}}}(\mu)\right)}{\mathcal{L}\left(0, \hat{\hat{\boldsymbol{\theta}}}(0)\right)}, & \hat{\mu} < 0, \\ -2\ln \dfrac{\mathcal{L}\left(\mu, \hat{\hat{\boldsymbol{\theta}}}(\mu)\right)}{\mathcal{L}\left(\hat{\mu}, \hat{\boldsymbol{\theta}}\right)}, & 0 \le \hat{\mu} \le \mu, \\ 0, & \hat{\mu} > \mu. \end{cases} \tag{6.7}$$

For $\hat{\mu} \geq \mu$, $\tilde{q}$ is set to 0, so that the tested μ is not placed in the rejection region for an upper limit. The test statistic is defined such that the larger value of $\tilde{t}_\mu$ correspond to greater disagreement between the data and the hypothesis value of μ.

To quantify the level of disagreement, we check the p-value,

$$p_\mu = \int_{\tilde{q}_{\mu,\text{obs}}}^{\infty} f(\tilde{q}_\mu \mid \mu)\, d\tilde{q}_\mu, \tag{6.8}$$

where $f(\tilde{q}_\mu \mid \mu)$ is the pdf of $\tilde{q}$, under the assumption that the tested μ is true.

Since larger values of $\tilde{q}_\mu$ correspond to greater incompatibility between the data and the tested signal hypothesis, p_μ represents the probability, assuming the hypothesis μ is true, of obtaining data that are at least as incompatible with that hypothesis as the observed data. Small p_μ therefore indicates that the tested signal strength μ is disfavored by the data. The relation between the p-value and $\tilde{q}_\mu$ can be found in Fig. 6.1.

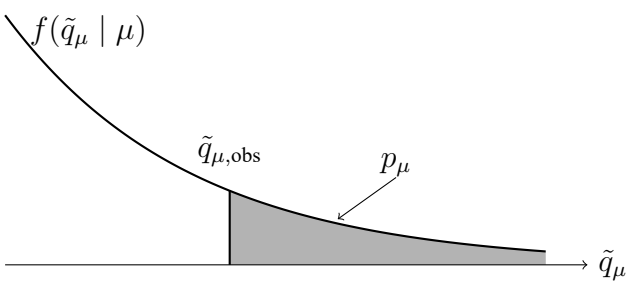


Figure 6.1: Illustration of the p-value for a tested signal strength μ. The shaded region represents the probability, under the hypothesis μ, of obtaining a value of the test statistic $\tilde{q}_\mu$ at least as large as the observed value $\tilde{q}_{\mu,\text{obs}}$.

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6.1.6 The 95% Confidence Level

Although the p-value quantify the incompatibility, to prevent excluding the signal before the experiment is sensitive enough to distinguish it from background, the CL_s ratio is used instead to derive the upper limit on the signal strength.

CL_s is defined as

$$CL_s \equiv \frac{CL_{s+b}}{CL_b} = \frac{1 - \Phi\left(\sqrt{\tilde{q}_\mu}\right)}{\Phi\left(\sqrt{\tilde{q}_{\mu,A}} - \sqrt{\tilde{q}_\mu}\right)} = \alpha, \tag{6.9}$$

and the 95% confidence level upper limit on the signal strength is obtained by finding the value μ for which $CL_s = 0.05$. Signal strength hypothesis with $CL_s < 0.05$ are excluded at 95% confidence level.

[explain the equation]

6.1.7 Expected Mass Exclusion

Within the asymptotic approximation, the expected limit can be obtained using the background-only Asimov dataset. The Asimov dataset is an artificial representative dataset, constructed based on the background model that was fitted using the 2017 SinglePhoton data with the signal strength set to zero and the nuisance parameters set to their nominal values. The resulting limit represents the upper limit that would be obtained in an ensemble of background-only experiment.

The upper limit on the signal strength can be converted into and upper limit on the

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signal production cross section times the branching fraction

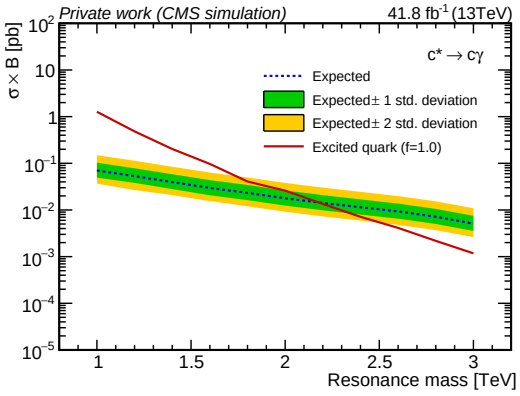
$$\sigma\mathcal{B} = \mu \times \sigma_{\text{theory}}\mathcal{B} \tag{6.10}$$



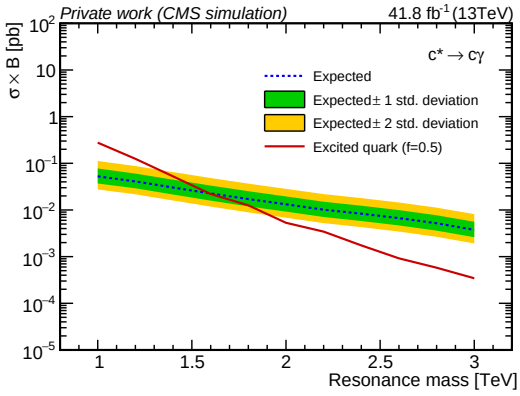
6.2 Results

The expected upper limits are illustrated in Fig 6.2. The expected 95% confidence level upper limits on $\sigma_{\text{theory}} \times \mathcal{B}$ are evaluated for each c^* mass hypothesis. A mass hypothesis is excluded when the theoretical cross section is larger than the 95% CL upper limit. The intersection of the theoretical prediction with the expected limit curve therefore determines the lower exclusion bound on M_{c^*}, which are found to be 2.18 TeV and 1.59 TeV for $f = 1.0$ and 0.5 respectively. For $f = 0.1$, the experiment is not sensitive enough to set a limit.

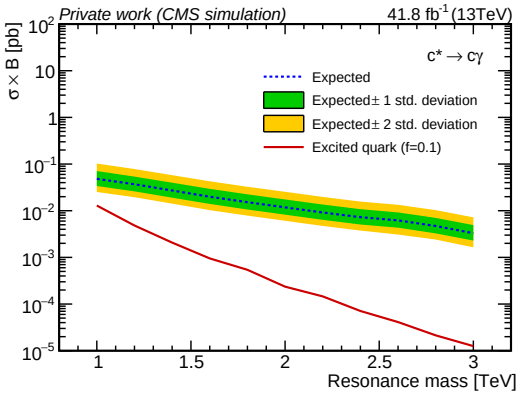
doi:10.5566/NTU2018XXXXX



(a) $f = 1.0$



(b) $f = 0.5$



(c) $f = 0.1$

Figure 6.2: Expected 95% C.L. upper limits on the production cross section times branching fraction for excited charm-quark production for different coupling strengths.

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Appendix A — DSCB fit for the signals

A.1 Introduction

A.2 Further Introduction

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Appendix B — Pythia Configuration

B.1 Introduction

B.2 Further Introduction

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